A PROOFS

A.1 Proof of Lemma 4

This conclusion is proved by constructing Π in an unnamed series calculus query $q : I[\![T, u_1, u_2, \ldots, u_n \mid \Pi]\!]$ and validating consistent mappings between $q(\mathbf{I})$ and $e(\mathbf{I})$. Consider the queried instance set $\{I_1, \ldots, I_m\}$ by e of schema $\{S_1, \ldots, S_m\}$, respectively. Initially, $\gamma = (T \in adom_T(\mathbf{I}(S_1 \ldots, S_m)))$. Formula Π_e is constructed by the following cases recursively. (1) When the subexpression e_x of e is I_k, formula $\Pi_x = S_k(T, A_1^k, \ldots, A_n^k)$; (2) When e_x is $\sigma_\theta(e_y)$, formula $\Pi_x = \Pi_y \wedge \theta$; (3) When e_x is $\pi_{A_1,\ldots,A_m}(e_y)$, formula

$$\begin{aligned}\Pi_x = &\exists v_{A1}, \ldots, v_{Am}((u_1 = v_{A1} \wedge \cdots \wedge u_m = v_{Am}) \wedge \\ &\exists v_{j_1}, v_{j_2}, \ldots, v_{j_l} \Pi_y(T, v_1, \ldots, v_{arity(\Pi_y)})),\end{aligned}$$

where $\{T, v_1, \ldots v_{arity(\Pi_y)}\}$ is the set of free variables in Π_y that is equal to $\{T\} \cup \{v_{A1}, \ldots, v_{Am}\} \cup \{v_{j_1}, \ldots, v_{j_l}\}$; (4) When e_x is $e_y \circ e_z$, formula $\Pi_x = \Pi_y \vee \Pi_z$; (5) When e_x is $e_y \bowtie e_z$, formula $\Pi_x = \Pi_y \wedge \Pi_z$; (6) When e_x is $e_y - e_z$, formula $\Pi_x = \Pi_y \wedge \neg\Pi_z$; (7) When e_x is $\pi_f(e_y)$, formula $\Pi_x = \exists v_{A1}, \ldots, v_{Am}(f(u, T, v_{A1}, \ldots, v_{Am}) \wedge \exists v_{j_1}, v_{j_2}, \ldots, v_{j_l} \Pi_y(T, v_1, \ldots, v_{arity(\Pi_y)}))$, where u, v are similar to case (3), supposing f has an arity $m + 2$. Finally, $\Pi = \Pi_e \wedge \gamma$ by including the timestamp predicate.

A.2 Proof of Lemma 5

Given any query $\{T, u_1, \ldots, u_m \mid \gamma \wedge \varphi\}$, it is possible to construct a legal expression e on instances $\mathbf{I}$ for equal mappings. For φ, let $\mathbf{I}(\varphi)$ denote the set of queried instances by $S(T, A_1, \ldots, A_n)$.

(1) When $\varphi = \psi(T, \mathbf{u}, \mathbf{v}) \wedge \xi(T, \ldots)$, we construct expression $e_\varphi = \pi_{A_1,\ldots A_m}(e_\psi \bowtie e_\xi)$;

(2) When $\varphi = \psi(T, \mathbf{u}, \mathbf{v}) \wedge \xi(\mathbf{u}', \mathbf{v}')$, the following expression generate equal mappings $e_\varphi = \pi_{A_1,\ldots A_m}(\sigma_\xi(e_\varphi))$;

(3) When $\varphi = \psi(T, \mathbf{u}, \mathbf{v}) \wedge \neg\xi(T, \ldots)$, the expression $\pi_{A_1,\ldots A_m}(e_\psi - e_\xi)$ generate the same evaluations;

(4) When $\varphi = \psi(T, \mathbf{u}, \mathbf{v}) \vee \xi(T, \ldots)$, we have, $e_\varphi = \pi_{A_1,\ldots A_m}(e_\psi \circ e_\xi)$;

(5) When $\varphi = \psi(T, \mathbf{u}, \mathbf{v}) \wedge \forall \mathbf{v}(\xi(\mathbf{u}, \mathbf{v}))$, expression $e_\varphi = \sigma_\xi(e_\psi \bowtie (\mathbf{I}(S_1) \circ \cdots \circ \mathbf{I}(S_k)))$ by comparing conditions ξ with the domain; likewise, e_φ can be constructed on $\exists \mathbf{v}(\xi)$.

For γ, it has timestamp definition $T \in adom_T(\mathbf{I}(S_1, \ldots, S_m))$ and timestamp/position predicates $\theta(T, i)$. When $\mathbf{I}(\varphi) \subseteq \{I_1, \ldots, I_m\}$, expression $e = \sigma_{\theta(T,i)}(e_\varphi)$; otherwise, it means that the result should not contain timestamps only existing in $\{I_1, \ldots, I_m\} \setminus \mathbf{I}(\varphi)$. Since the valuations are assigned by each timestamp, the formula $e = \pi_{T,\mathbf{u}_\varphi}(\sigma_{\theta(T,i) \wedge (\mathbf{u}_\varphi)}(e_\varphi \circ (I_{v_1} \circ \cdots \circ I_{v_l})))$, where $\{I_1, \ldots, I_m\} = \{I_{v_1}, \ldots, I_{v_l}\} \cup \mathbf{I}(\varphi)$, $\mathbf{u}_\varphi$ is the set of projected attributes in φ, and predicate $\mathbf{u}_\varphi$ means all attributes not being empty simultaneously.